

Project Title:

Azure File Sync with an On-Premises File Server

Team Number: T175

Project Number: P073

Problem Statement:

Nowadays, a lot of companies save their files on local servers. But the more data there is, the more challenging it is to manage and store them correctly and keep backups. Our solution allows to resolve these challenges by connecting the local file server with the cloud storage (Microsoft Azure).

Abstract:

Modern cloud applications often generate a large number of outbound network connections, especially when interacting with external APIs, databases, and other cloud services. When the number of outbound connections increases significantly, limited Source Network Address Translation (SNAT) ports can become exhausted, resulting in connection failures, timeouts, and performance issues that may appear to be application-level problems.

This project focuses on understanding, implementing, and optimizing outbound connectivity using **Azure Load Balancer** and its outbound rules. The proposed system will analyze the outbound traffic generated by a chatty application and study how SNAT ports are allocated to different connections. An Azure-based environment will be designed with backend virtual machines, a Load Balancer, and appropriate outbound connectivity. Test traffic will then be generated to observe connection behavior and identify potential SNAT port exhaustion.

The project will also involve estimating the required outbound capacity based on traffic patterns and connection requirements. Different configurations will be analyzed to determine an appropriate SNAT capacity and reduce the possibility of port exhaustion. Monitoring and load testing will be used to evaluate the effectiveness of the proposed configuration under different traffic conditions. The final outcome will provide a practical understanding of Azure outbound networking and demonstrate how proper SNAT planning can improve the reliability and scalability of cloud applications.

Use Cases:

- File sync: You will be able to synchronize your local files with remote files in the cloud storage.
- Cloud storage: With our solution, you can store extra files in Azure.
- Cloud tiering: Sync your frequently used files locally while storing your seldom accessed files in the cloud.
- Multiple Offices: You can manage and synchronize all your files between different offices.
- Backup: You can make the additional copy of your local files on Azure.

Requirements:

- Azure subscription
- Windows Server/VM
- Azure Storage Account
- Azure File Share
- Storage Sync Service
- Azure File Sync agent
- Internet connect

Team members:

2400032195 – S. MADHURI DEVI

2400032747 – B. LAKSHMI SRAVANTHI

2400032991 - BAPATHU YASWANTH OBUL REDDY

2400032964 - VISHNUBHOTLA H S S N S RAMA Koushik